

Let, $\neg(p \vee q \vee r)$ be True. By De Morgan's law, $(\neg p \wedge \neg q \wedge \neg r)$ is True.
 By defn. of AND, $\neg p$ is True, $\neg q$ is True, $\neg r$ is True.

$\neg p$: x is not odd, i.e. x is even (Defn. 1)

$\neg q$: y is not odd, i.e. y is even (Defn. 1)

$\neg r$: z is not odd, i.e. z is even (Defn. 1)

By Defn. 1, $\exists$ integers k_1, k_2, k_3 such that $x = 2k_1, y = 2k_2, z = 2k_3$

$$\therefore x+y+z = 2k_1 + 2k_2 + 2k_3 = 2(k_1 + k_2 + k_3)$$

It's easy to see that $(k_1 + k_2 + k_3)$ is an integer. By defn 1, $x+y+z$ is even.

$\neg s$: $(x+y+z)$ is not odd, i.e. $(x+y+z)$ is even (Defn. 1)

$\therefore \neg s$ is True. By defn. of conditional, $\boxed{\neg(p \vee q \vee r) \rightarrow \neg s \text{ is True}}$

$\therefore s \rightarrow (p \vee q \vee r)$ is True

$\therefore$ If $x+y+z$ is odd, then atleast one of x, y or z is odd $\square$

17) Use a proof by contraposition to show that if $(x+y) \nmid 2$, where x and y are real numbers, then $x \nmid 1$ or $y \nmid 1$.

Soln: $p: (x+y) \nmid 2; x, y \in \mathbb{R}$ $q: x \nmid 1, x \in \mathbb{R}$ $r: y \nmid 1, y \in \mathbb{R}$

We shall prove $p \rightarrow (q \vee r)$ is True by contraposition.

$\therefore$ We shall prove $\neg(q \vee r) \rightarrow \neg p$ is True.

Let, $\neg(q \vee r)$ be True. By De Morgan's Law, $\neg q \wedge \neg r$ is True. By the defn. of AND, $\neg q$ is True and $\neg r$ is True.

$\neg q: x \nmid 1, x \in \mathbb{R}$ $\neg r: y \nmid 1, y \in \mathbb{R}$.

$\therefore x \nmid 1, y \nmid 1 \Rightarrow (x+y) \nmid 2$ is True. $\therefore \neg(x+y) \nmid 2$ is True

or, $\neg p$ is True. By defn. of conditional, $\neg(q \vee r) \rightarrow \neg p$ is True.